

AttnGAN: Fine-Grained Text-to-Image Generation

CS 5031 · APPLIED COMPUTER VISION · SPRING 2026

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Task Definition

Given a natural language bird description — e.g., *"A red bird with a short black beak and white belly"* — the model generates a **photorealistic image** at 64×64, 128×128, or 256×256 resolution.

Unlike earlier GAN methods conditioned on a single sentence vector, AttnGAN introduces **word-level attention**: dynamically attending to individual words while refining specific image regions.

- ❶ DAMSM loss enforces semantic alignment between generated pixels and text tokens — a core contribution of Xu et al. (CVPR 2018).



Dataset: CUB-200-2011

Property	Value
Dataset	CUB-200-2011 (Caltech-UCSD Birds)
Training images	8,855 × 10 captions = 88,550 pairs
Testing images	2,933 × 10 captions = 29,330 pairs
Bird species	200
Label type	Free-form text captions
Image format	JPEG, variable resolution, RGB

Each image has **10 independently written captions**. One is sampled randomly per epoch, acting as implicit data augmentation. Varied poses, backgrounds, and lighting make fine-grained detail generation particularly challenging.



Figure 1. Example results of the proposed AttnGAN. The first row

Data Pre-processing Pipeline

1

Text: Tokenize

Split to words, lowercase, map to vocabulary IDs via shared word embedding (dim=300).

2

Text: Pad / Truncate

Fixed length = 18 tokens (cfg.TEXT.WORDS_NUM). Actual lengths tracked for `pack_padded_sequence` in bi-LSTM.

3

Images: AttnGAN Train

Resize $\rightarrow$ 76 $\times$ 76, Random crop $\rightarrow$ 64 $\times$ 64, H-flip, Normalize $[-1,1] \rightarrow 3 \times 64 \times 64$. Bilinear upsample to 128 $\times$ 128 (and 256 $\times$ 256 on 8GB+).

4

Images: DAMSM

Resize $\rightarrow$ 356 $\times$ 356, Random/Center crop $\rightarrow$ 299 $\times$ 299, Normalize $[-1,1] \rightarrow 3 \times 299 \times 299$. Wrong pairs formed by cyclic batch shift.

Network Architecture

Component	Role	Key Details
RNN_ENCODER	Text encoding	word_emb(vocab,300) → bi-LSTM(hidden=256) → word feats (B×256×T) + sent emb (B×256)
CNN_ENCODER	Image encoding (DAMSM)	Inception-v3 (frozen) → local B×256×289 + global B×256 via learnable projections
CA_NET	Conditioning Augmentation	$\bar{e} \rightarrow \text{MLP} \rightarrow \mu, \log \sigma$; sample $c \sim N(\mu, \sigma^2)$; KL regularization
G0 / G1 / G2	Generator stages	64×64 → 128×128 → 256×256 (8GB+ only); z(100)+c(128), ResBlocks, Tanh
Attention Module	Word-region alignment	$\beta_{j,i} = \text{softmax}(e_i \cdot v_j)$; context $c_j = \sum \beta_{j,i} e_i$
D_NET64/128	Discriminators	Spectral-norm convs → 4×4 feat → unconditional + conditional heads (with sent_emb)

📌 Our addition: **spectral normalization** on all discriminator convolutions (not in the original paper) — stabilizes training and prevents gradient explosion.

Loss Functions

Five-Term Objective (losses.py)

Term	Weight
L_D — Discriminator (per stage)	—
L_G — Generator adversarial (per stage)	1.0
L_KL — KL vs N(0,I)	1.0
L_DAMSM — $L_1^w + L_2^w + L_1^s + L_2^s$ (word + sentence, both directions)	$\lambda = 5$
L_total (G objective)	$\sum L_{\{G,i\}} + L_{KL} + 5 \cdot L_{DAMSM}$

Same-class pairs in DAMSM are masked as non-negatives to avoid false penalties. $\lambda = 5$ for CUB ($\lambda = 50$ for COCO).

Discriminator Loss Detail

Each stage D minimizes:

$$\begin{aligned} \mathcal{L}_D = & 0.5 \cdot \text{BCE}(D_{\text{unc}}(\text{real}), 1) + 0.5 \cdot \text{BCE}(D_{\text{unc}}(\text{fake}), 0) \\ & + 0.5 \cdot \text{BCE}(D_{\text{cond}}(\text{real}, t), 1) + 0.25 \cdot \text{BCE}(D_{\text{cond}}(\text{fake}, t), 0) \\ & + 0.25 \cdot \text{BCE}(D_{\text{cond}}(\text{wrong}, t), 0) \end{aligned}$$

The wrong-pair term penalizes matching real images to mismatched captions — critical for semantic grounding.

Hyperparameters & GPU-Adaptive Configuration

Hyperparameter	Value
Optimizer	Adam
LR (G, D, encoders)	0.0002
Adam β_1 / β_2	0.5 / 0.999
Z_DIM / Embed dim	100 / 256
Word embed dim	300
ResBlocks / stage	2
λ (DAMSM weight)	5.0 (CUB)
AttnGAN epochs	600
DAMSM epochs	120 
LR scheduler (DAMSM)	StepLR, step=100, γ =0.5

GPU Tier Auto-Selection

Tier	Batch (GAN)	GF_DIM	Stages
4 GB	2	16	2
8 GB	8	32	3
16 GB	16	64	3
CPU	2	16	2



Mixed Precision (AMP) enabled everywhere: ~40% memory reduction, ~30% faster training. Gradient checkpointing saves ~400 MB on 4GB GPUs.

Our Implementation Improvements



Dynamic GPU Tier Detection

VRAM measured at startup → 4 tiers; batch size, dims, and stages auto-configured. Same codebase runs on 4GB to 16GB or CPU without manual edits.



Spectral Normalization

All discriminator convolutions use `spectral_norm` (absent from original paper). Prevents gradient explosion and reduces mode collapse.



Mixed Precision (AMP)

`torch.cuda.amp` autocast + GradScaler throughout both training scripts. ~40% memory reduction and ~30% faster wall-clock training.

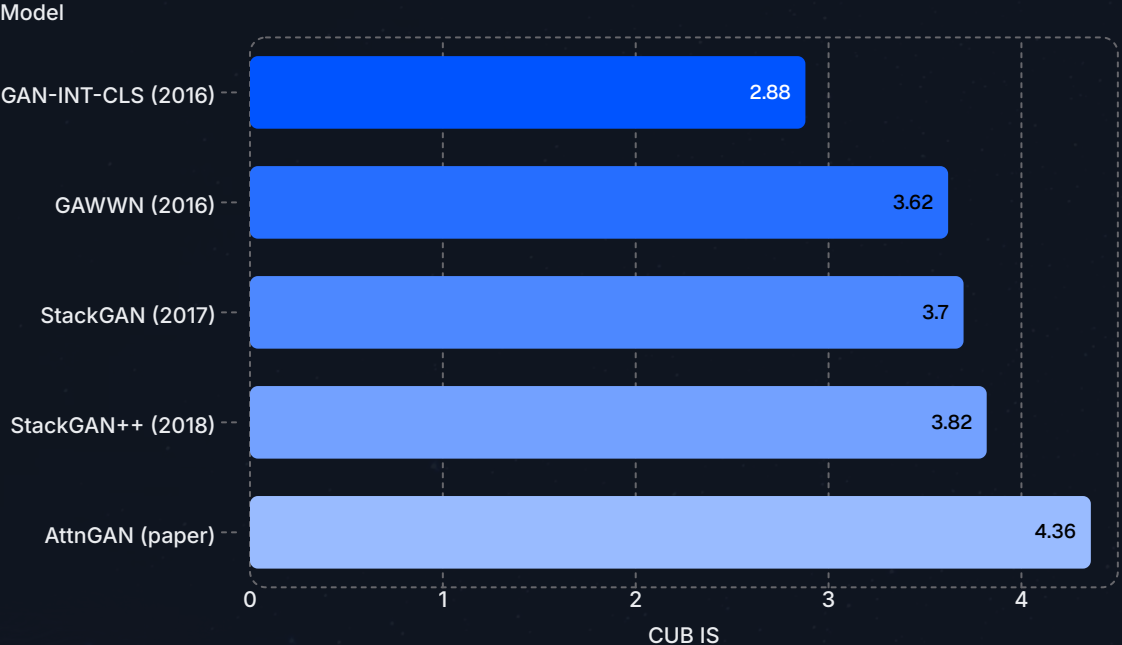


Gradient Checkpointing

`torch.utils.checkpoint` on G_NET stages (4GB tier). Frees ~400 MB of activation memory, enabling training where it would otherwise OOM.

SOTA Comparison & Training Progress

Published Inception Score on CUB (↑ better)



Our Training Status

DAMSM Pretraining ✓

120/120 epochs complete. Loss reduced **15.06 → 8.12** (46% reduction), confirming strong semantic embedding.

AttnGAN Training ↻

21/600 epochs at submission. D loss: **1.384 → 1.208** (near $\ln(4)$ equilibrium). G loss: 4.8–5.2 (expected variance at batch=2). No mode collapse detected.

ⓘ Full IS/FID evaluation requires ~600 epochs. Early diagnostics confirm healthy GAN convergence.

Early Samples & Monitoring

Generated Samples — Epoch 1

Basic color structure and coarse blob-like shapes emerge at epoch 1. Distinct bird shapes, textures, and species-specific features are expected by epochs 50–100 as the attention mechanism begins to focus on discriminative word-region alignments.

Live diagnostics via **TensorBoard**: step/epoch losses for DAMSM and AttnGAN, generated image grids every snapshot interval, and GPU utilization tracking.

References

[1] Xu et al. (2018). <i>AttnGAN: Fine-Grained Text to Image Generation</i> . CVPR 2018. arXiv:1711.10485	[2] Zhang et al. (2017). <i>StackGAN: Text to Photo-realistic Image Synthesis</i> . ICCV 2017.
[3] Reed et al. (2016). <i>Generative Adversarial Text to Image Synthesis</i> . ICML 2016.	[4] Wah et al. (2011). <i>The Caltech-UCSD Birds-200-2011 Dataset</i> .
[5] Goodfellow et al. (2014). <i>Generative Adversarial Nets</i> . NeurIPS 2014.	[6] Miyato et al. (2018). <i>Spectral Normalization for GANs</i> . ICLR 2018.

64×64 Stage

Color blobs with rough spatial structure

128×128 Stage

Upsampled; fine detail still forming